

Pranav Mishra

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Education

Master of Science, Computer Science [Graduate Assistant]

Aug 2023 - May 2025

University of Illinois at Chicago, Illinois, USA

GPA: 3.6/4.0

Coursework: Advanced ML, Applied AI, Advanced NLP, Computer Vision, VR, Game Design, Object-Oriented Programming

Bachelor of Science, Computer Science and Engineering

Aug 2019 - June 2023

Dayananda Sagar College of Engineering, Bangalore, Karnataka, India

GPA: 4.0/4.0

Coursework: AI & ML, Software Engineering, Computer Networks, Operating Systems, Data Structures & Algorithms, DBMS

Technical Experience

AI/ML Intern | WheelPrice ↗ | Charlotte, NC

July 2025 - Jan 2026

- Building end-to-end ML prototype for automotive part fitment prediction using PyTorch and computer vision models
- Deployed production-grade CMS blog system using React TypeScript with Node.js Express backend, MongoDB database, and RESTful APIs, scaling daily webapp viewership by 10-20k through enhanced content delivery and SEO implementation

Research Software Engineer | UIC- VARE Lab ↗ | Chicago, IL

Feb 2024 - May 2026*

- Built virtual patient system using Unreal Engine and C++. Deployed RESTful APIs with Python backend for data analysis ↗
- Deployed end-to-end audio ML pipeline with Flask backend comparing MFCC feature engineering, CNN architectures, and Transformer models achieving 98.52% accuracy; deployed real-time inference system with VAD & robustness testing. ↗

Research & Publications

TeamMedAgents: Medical Decision Making with SLMs through Teamwork

GitHub | Paper | SLMs-v2.0

- Engineered modular multi-agent system using Google ADK with prompt engineering for medical QA across 8 benchmarks
- Achieved 77.63% accuracy on LLMs and 3.1x inference speedup on 4B models through collaborative deliberation, trust-weighted voting, and structured multi-turn reasoning with comprehensive evaluation frameworks

MetaRAG: Metadata Enrichment Framework for Enterprise RAG Systems

GitHub | IEEE Paper

- Engineered metadata enrichment pipeline using LLMs with advanced embeddings achieving 82.5% precision, 0.925 Hit Rate@10, and 25% hallucination reduction through cross-encoder reranking with comprehensive evaluation framework
- Deployed full-stack system on AWS with CI/CD pipelines, vector clustering, evaluation metrics, and performance monitoring

Projects

MockFlow-AI: Real-Time Voice Interview Platform with Multi-Agent Architecture

GitHub | Website

- Engineered production full-stack LLM application with sub-400ms latency using React TypeScript frontend, Node.js backend, and multi-agent orchestration with streaming STT-LLM-TTS pipelines for real-time conversational AI.
- Shipped end-to-end features like RAG-based retrieval, feedback loops, performance monitoring, & cloud-native deployment

SnakeAI-MLOps: Multi-Agent Reinforcement Learning Snake Game

GitHub | Demo

- Engineered RL training framework in C++ with LibTorch, PyTorch integration, and CUDA GPU acceleration.
- Built production MLOps pipeline with model comparison tools, Docker CI/CD, and real-time gameplay integration

MedRAG Avatar Platform - Intelligent Healthcare Conversational AI System

GitHub | Demo

- Architected production RAG system with React TypeScript frontend, FastAPI RESTful backend, Azure Cosmos DB, and Pinecone vector search achieving 95% accuracy for context-aware conversational avatars with source attribution

Skills

TECHNICAL SKILLS: TypeScript, JavaScript, React, Node.js, Next.js, Python, C++, C#, FastAPI, Django, PostgreSQL, MongoDB, Docker, Git, RESTful APIs, TensorFlow, PyTorch, LangChain, AWS (Lambda, Bedrock, S3), Azure, GCP, DevOps

APPLICATIONS: LLM Applications, RAG Systems, Agent Workflows, Full-Stack Development, MLOps, Cloud Computing, DevOps, Agile Development, Performance Monitoring, Model Evaluation, Vector Databases, Embedding Pipelines, API Design

Extracurricular

Winner of MIT XR Hackathon | Built Meta Quest 3 app using Unity and Hugging Face for spatial data visualization ↗

INFORMS Analytics+ Presented MetaRAG research to 700+ professionals | First place HINT 5.0 Web3 virtual museum ↗